

ITEC 3100 – Introduction to Networks

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Lab 1: *(Use a different color font for your written answers)*

Q1. In your own words list **what each hardware component does** and also insert a picture of it:

- a. CPU-The brains, determines the price of the computer and needs other components.



- b. RAM- Stores everything



- c. HDD- Magnetic hard drive that stores data and is more reliable than SSDs



- d. SSD- Solid state drive that stores data and is replacing HDDs due to speed



- e. M.2 drive- An SSD that plugs into the motherboard



- f. Motherboard- The main circuit of a computer that allows communication between components



g. Server- Provides for devices over a network



Q2. What are the different families of Intel i-Series CPUs?
Which one is the recommended one today for an IT student laptop? I3,i5,i7,i9/9x, An IT student laptop is recommended a i7

Q3. What are the primary differences between the families of CPUs? Cores and cache

Q4. What is a benefit of an i5 over and i7 other than price?
Better heat management since it's not as strong

Q5. What is the difference between a mechanical HDD and a SSD? What are the advantages and disadvantages of each?

Mechanical HDD is more reliable because more information is known since SSDs haven't been around as long

SSD Advantages – speed, quiet, durable

SSD Disadvantages- expensive

HDD Advantages- Bigger Storage, cheaper, reliable

HDD Disadvantages- Loud, Size, power use

Q6. What are the newest and fastest type of SSD drives available and its benefits? M.2 drives, offers fast speed, low latency and handles data well

Q7. What is the recommended amount of RAM in a system today and why? 16 GB because the RAM stores everything so with 16 GB you can comfortably store data and run programs without any storage issues.

Q8. How fast (Ghz) should your CPU be able to run today? What is the purpose of the Turbo feature? > 4.0 GHZ, The purpose is to increase the processor average speed

Q9. What is a bit and how many bits make a Byte?
A bit 0,1,
8 bits make a byte

Q10. Which operating system is the most popular today and why? [Windows, %70 percent of the world uses it.](#)

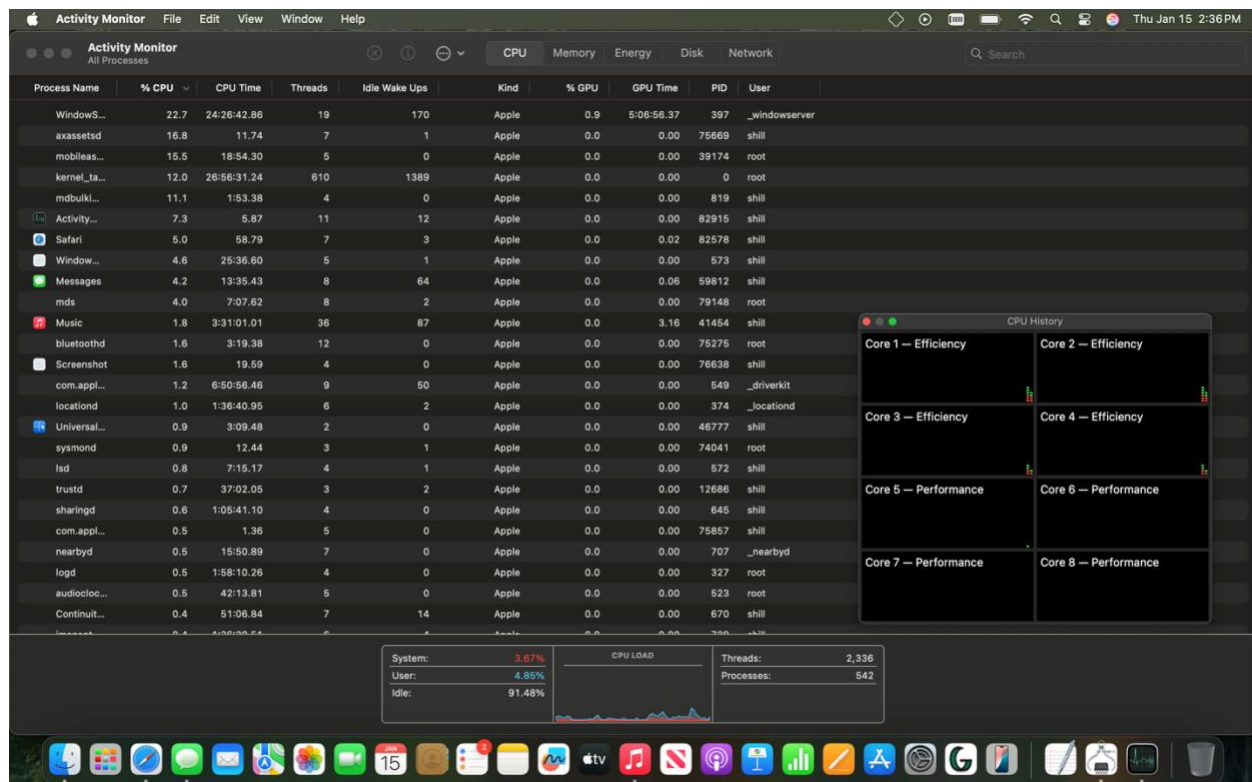
Q11. What is a typical size of a non-HD movie? How many GB make up a TB? [5 GB, 1000 GB= 1 TB](#)

Q12. What are the 4 basic functions of a computer?
[Input, Processing, Output, and Storage](#)

Q13. Do some research to find out what your CPU is and find out the following: [Apple M1 CPU](#)

- Base speed [Around 2.06 GHz](#)
- Max speed [3.22 GHz](#)
- Cores [8](#)
- Threads [8](#)
- Cache [26.6](#)

Insert screenshot of Task Manager (Windows) or System/Activity Monitor (Mac) that we took in class.



Complete and upload to the Chapter 1 Lab dropbox.